CSE563: APPLIED MATHEMATICS

Question Bank with Solution Key • Session 2026-27

Unit I: Basic Mathematics in Computer Science

Maps to CO1: Recall the basic concepts of mathematics in computer science and engineering.

Q1. State the value of gcd(84, 30) using the Euclidean algorithm, and compute lcm(84, 30).

Syllabus: Number Theory | CO1 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

84 = 2(30) + 24 • 30 = 1(24) + 6 • 24 = 4(6) + 0. Thus, **gcd(84, 30) = 6**.

lcm(84, 30) = (84 × 30) / gcd(84, 30) = 2520 / 6 = **420**.

Q2. Illustrate the inclusion-exclusion principle by finding how many integers from 1 to 150 are divisible by 3 or by 7.

Syllabus: Set Theory | CO1 | Bloom L2 | Difficulty: Easy | Start word: Illustrate

Solution:

$|A| = \lfloor 150/3 \rfloor = 50$, $|B| = \lfloor 150/7 \rfloor = 21$, $|A \cap B| = \lfloor 150/21 \rfloor = 7$.

$|A \cup B| = |A| + |B| - |A \cap B| = 50 + 21 - 7 =$ **64**.

Q3. Calculate the truth value of (p ∨ q) → r for all truth assignments, and state in how many rows the expression evaluates to True.

Syllabus: Logic | CO1 | Bloom L3 | Difficulty: Medium | Start word: Calculate

Solution:

(p ∨ q) is False only when p=F, q=F (2 rows: FFF, FFT → both give True for implication).

For the remaining 6 rows where (p ∨ q) = True: evaluated as True when r=T (3 rows: TTT, TFT, FTT) and False when r=F (3 rows: TTF, TFF, FTF).

Total True rows = 2 + 3 = **5 rows out of 8**.

Q4. Determine the number of equivalence classes of congruence modulo 5 on ℤ, and list the four smallest non-negative members of the class containing 17.

Syllabus: Functions and Relations | CO1 | Bloom L3 | Difficulty: Easy | Start word: Determine

Solution:

Congruence modulo 5 produces **5 equivalence classes**: [0], [1], [2], [3], [4].

17 ≡ 2 (mod 5), so 17 ∈ [2]. The 4 smallest non-negative members are **2, 7, 12, 17**.

Q5. Apply calculus rules to solve the following:

Syllabus: Calculus | CO1 | Bloom L3 | Difficulty: Medium | Start word: Apply

Solution:

(a) For $y = 4x^3 - 6x^2 + 9$: $dy/dx =$ **$12x^2 - 12x$**.

(b) For $f(x, y) = 3x^2y^2 + 5xy^3$: $\partial f/\partial x =$ **$6xy^2 + 5y^3$**, $\partial f/\partial y =$ **$6x^2y + 15xy^2$**.

(c) At (1, 2): $\partial f/\partial x = 6(1)(4) + 5(8) = 24 + 40 =$ **64**; $\partial f/\partial y = 6(1)(2) + 15(1)(4) = 12 + 60 =$ **72**.

(d) Evaluate $\int_0^2 (3x^2 + 4x - 1)\,dx = [x^3 + 2x^2 - x]_0^2 = (8 + 8 - 2) - 0 =$ **14**.

Q6. Examine modular exponentiation and inverses under modulo 11.

Syllabus: Number Theory | CO1 | Bloom L4 | Difficulty: Medium | Start word: Examine

Solution:

(a) $2^5 \bmod 11 = 32 \bmod 11 =$ **10**.

(b) $4^4 \bmod 11 = 256 = 23(11) + 3 =$ **3**.

(c) Remainder of 3^{10} divided by 11: By Fermat's Little Theorem, $3^{10} \equiv$ **1 (mod 11)**.

(d) Find $x \in \{0,...,10\}$ such that $4x \equiv 1 \pmod{11}$: $4(3) = 12 \equiv 1 \pmod{11}$, so **x = 3**.

Q7. Evaluate the Boolean function F(A, B, C) = ∑ m(1, 3, 4, 5, 6, 7) using a 3-variable Karnaugh Map.

Syllabus: Boolean Algebra | CO1 | Bloom L5 | Difficulty: Difficult | Start word: Evaluate

Solution:

(a) Binary minterms: m1=001, m3=011, m4=100, m5=101, m6=110, m7=111.

(b) Map grouping: Group 1 (m4, m5, m6, m7) gives **A**. Group 2 (m1, m3, m5, m7) gives **C**.

(c) Minimal SOP expression: **F = A + C**.

(d) Literal reduction: Canonical form has 6 × 3 = 18 literals; Minimal form has 2 literals. Total saving = **16 literals**.

Q8. Construct a verification of the sum of squares formula $\sum_{i=1}^{n} i^2 = [n(n + 1)(2n + 1)] / 6$.

Syllabus: Mathematical Induction | CO1 | Bloom L6 | Difficulty: Difficult | Start word: Construct

Solution:

(a) Verification for n=1, 2, 3:

• n=1: LHS=1, RHS=(1 × 2 × 3)/6 = 1 • n=2: LHS=1+4=5, RHS=(2 × 3 × 5)/6 = 5 • n=3: LHS=1+4+9=14, RHS=(3 × 4 × 7)/6 = 14.

(b) For n=4: LHS = 1 + 4 + 9 + 16 = 30; RHS = (4 × 5 × 9)/6 = 30.

(c) For n=10: RHS = (10 × 11 × 21)/6 = **385**.

(d) Value of n where $\sum i^2 = 55$: $(n(n+1)(2n+1))/6 = 55 \Rightarrow n(n+1)(2n+1) = 330 \Rightarrow$ **n = 5** (since 5 × 6 × 11 = 330).

Unit II: Vectors and Matrices

Maps to CO2: Demonstrate the use of vectors and matrices in image transformations.

Q9. State the dot product u • v for u = (3, −2, 5) and v = (4, 1, −2).

Syllabus: Vector and Operations | CO2 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

u • v = (3)(4) + (−2)(1) + (5)(−2) = 12 − 2 − 10 = **0** (Vectors are orthogonal).

Q10. State the Euclidean norm of w = (2, 6, 9) and the distance between points P(2, −1) and Q(7, 11).

Syllabus: Vector and Operations | CO2 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

$||w|| = \sqrt{(4 + 36 + 81)} = \sqrt{121} =$ **11**.

$d(P, Q) = \sqrt{((7 - 2)^2 + (11 - (-1))^2)} = \sqrt{(25 + 144)} = \sqrt{169} =$ **13**.

Q11. Illustrate the cross product by evaluating (1, 3, 4) × (2, 5, 1).

Syllabus: Vector and Operations | CO2 | Bloom L2 | Difficulty: Medium | Start word: Illustrate

Solution:

(1, 3, 4) × (2, 5, 1) = (3(1) − 4(5), 4(2) − 1(1), 1(5) − 3(2)) = **(−17, 7, −1)**.

Check orthogonality: (−17)(1) + 7(3) + (−1)(4) = −17 + 21 − 4 = 0.

Q12. Illustrate determinant and inverse evaluation for matrix A = [[5, 2], [3, 4]].

Syllabus: Matrices and Operations | CO2 | Bloom L2 | Difficulty: Easy | Start word: Illustrate

> **Solution:**
> det(A) = (5)(4) − (2)(3) = 20 − 6 = **14**.
> A^{-1} = (1/14) * [[4, −2], [−3, 5]] = **[[2/7, −1/7], [−3/14, 5/14]]**.

Q13. Determine the rank and nullity of matrix M = [[1, 3, 2], [3, 9, 6]].

Syllabus: Matrices and Operations | CO2 | Bloom L3 | Difficulty: Easy | Start word: Determine

> **Solution:**
> Row 2 = 3 × Row 1. Thus, only 1 linearly independent row ⇒ **Rank = 1**.
> By Rank-Nullity Theorem: Nullity = Columns − Rank = 3 − 1 = **2**.

Q14. Calculate the eigenvalues of matrix A = [[5, 2], [2, 2]].

Syllabus: Matrices and Operations | CO2 | Bloom L3 | Difficulty: Medium | Start word: Calculate

> **Solution:**
> det(A − λI) = (5 − λ)(2 − λ) − 4 = λ^2 − 7λ + 6 = 0.
> (λ − 6)(λ − 1) = 0 ⇒ Eigenvalues are $\boldsymbol{\lambda_1 = 6, \lambda_2 = 1}$.

Q15. Compute the pixel value at position (0.4, 0.6) using bilinear interpolation, given surrounding values f(0,0)=12, f(1,0)=24, f(0,1)=36, f(1,1)=48.

Syllabus: Image Interpolation | CO2 | Bloom L3 | Difficulty: Medium | Start word: Compute

> **Solution:**
> Along x: f(0.4, 0) = 12(0.6) + 24(0.4) = 7.2 + 9.6 = 16.8 • f(0.4, 1) = 36(0.6) + 48(0.4) = 21.6 + 19.2 = 40.8.
> Along y: f(0.4, 0.6) = 16.8(0.4) + 40.8(0.6) = 6.72 + 24.48 = **31.2**.
> Nearest-neighbour value: (0.4, 0.6) rounds to (0, 1), giving **36**.

Q16. Analyze a PCA outcome where the covariance matrix eigenvalues are 9, 4, and 3. Compute variance explained by PC1 and by (PC1 + PC2).

Syllabus: Principal Component Analysis | CO2 | Bloom L4 | Difficulty: Medium | Start word: Analyze

> **Solution:**
> Total Variance = 9 + 4 + 3 = 16.
> PC1 variance ratio = 9 / 16 = **56.25%**.
> PC1 + PC2 variance ratio = (9 + 4) / 16 = 13 / 16 = **81.25%**.

Q17. Solve the following linear system using Gaussian elimination: x + 2y + z = 8, 3y + 4z = 11, 2x + 5y − z = 9.

Syllabus: Matrices and Operations | CO2 | Bloom L3 | Difficulty: Medium | Start word: Solve

> **Solution:**
> Augmented matrix [[1, 2, 1 | 8], [0, 3, 4 | 11], [2, 5, −1 | 9]].
> R3 → R3 − 2R1 gives [[0, 1, −3 | −7]].
> R3 → R3 − (1/3)R2 gives [[0, 0, −13/3 | −32/3]] ⇒ z = 32/13.
> Back-substitution yields: 3y + 4(32/13) = 11 ⇒ y = 5/13. Then x + 2(5/13) + 32/13 = 8 ⇒ **x = 62/13, y = 5/13, z = 32/13**.

Q18. Examine the dataset (2, 2), (4, 6), (6, 4), (8, 8) by Principal Component Analysis.

Syllabus: Principal Component Analysis | CO2 | Bloom L4 | Difficulty: Difficult | Start word: Examine

> **Solution:**
> Mean = (5, 5). Centered data: (−3, −3), (−1, 1), (1, −1), (3, 3).
> Covariance matrix C = (1/3) * [[20, 16], [16, 20]] = [[6.67, 5.33], [5.33, 6.67]].
> Eigenvalues: det(C − λI) ⇒ λ_1 = 12.0, λ_2 = 1.33.
> Variance explained by PC1 = 12.0 / 13.33 = **90.0%**.

Q20. Construct Fisher's Linear Discriminant direction for C1 = {(1, 1), (2, 3)} and C2 = {(5, 5), (6, 7)}.

Syllabus: Linear Discriminant Analysis | CO2 | Bloom L6 | Difficulty: Difficult | Start word: Construct

Solution:

m1 = (1.5, 2.0), m2 = (5.5, 6.0) ⇒ m2 − m1 = (4.0, 4.0).

Scatter matrices S1 = S2 = [[0.5, 1.0], [1.0, 2.0]], Sw = S1 + S2 = [[1.0, 2.0], [2.0, 4.0]].

Using pseudo-inverse / regularized Sw: Direction vector $\mathbf{w} \propto (1, 1)^T$.

Unit III: Probability and Statistics

Maps to CO3: Apply concepts of probability and statistics for inference and regression analysis.

Q21. State the probability of rolling a sum of 8 with two fair dice.

Syllabus: Basic Probability | CO3 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

Favorable pairs: (2,6), (3,5), (4,4), (5,3), (6,2) ⇒ 5 outcomes out of 36.

P(Sum = 8) = **5 / 36 ≈ 0.1389**.

Q22. State the mean, variance, and standard deviation of a Binomial variable with n = 16 and p = 0.25.

Syllabus: Probability Distribution | CO3 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

Mean μ = np = 16(0.25) = **4**.

Variance σ^2 = np(1−p) = 16(0.25)(0.75) = **3**.

Standard Deviation σ = √3 ≈ **1.732**.

Q23. Illustrate the Poisson distribution by computing P(X = 2) when λ = 3.

Syllabus: Probability Distribution | CO3 | Bloom L2 | Difficulty: Medium | Start word: Illustrate

Solution:

$P(X = k) = (e^{-\lambda} \lambda^k) / k! \Rightarrow P(X = 2) = (e^{-3} \times 3^2) / 2! = (0.049787 \times 9) / 2 =$ **0.2240**.

Q24. Illustrate the addition rule: P(A) = 0.5, P(B) = 0.4, P(A ∩ B) = 0.15. Compute P(A ∪ B) and check independence.

Syllabus: Basic Probability | CO3 | Bloom L2 | Difficulty: Easy | Start word: Illustrate

Solution:

P(A ∪ B) = 0.5 + 0.4 − 0.15 = **0.75**.

Check independence: P(A)P(B) = 0.5 × 0.4 = 0.20 ≠ P(A ∩ B) = 0.15. Thus, A and B are **dependent**.

Q25. Compute conditional probabilities P(A|B) and P(B|A) given P(A) = 0.5, P(B) = 0.4, P(A ∩ B) = 0.15.

Syllabus: Compound Probability | CO3 | Bloom L3 | Difficulty: Medium | Start word: Compute

Solution:

P(A|B) = P(A ∩ B) / P(B) = 0.15 / 0.4 = **0.375**.

P(B|A) = P(A ∩ B) / P(A) = 0.15 / 0.5 = **0.300**.

Q26. Calculate P(X > 115) for a normal distribution X ~ N(100, 10^2).

Syllabus: Probability Distribution | CO3 | Bloom L3 | Difficulty: Medium | Start word: Calculate

Solution:

z-score z = (115 − 100) / 10 = 1.5.

P(Z > 1.5) = 1 − P(Z ≤ 1.5) = 1 − 0.9332 = **0.0668**.

Q27. Determine standard error of the mean and 95% confidence margin of error for sample n = 100, s = 15.

Syllabus: Frequentist Inference | CO3 | Bloom L3 | Difficulty: Medium | Start word: Determine

Solution:

Q29. Analyze a Bayes' theorem defect problem: Factory X produces 70% of components (1% defective), Factory Y produces 30% (4% defective). An item picked is defective. Compute P(X|D).

Syllabus: Compound Probability | CO3 | Bloom L4 | Difficulty: Difficult | Start word: Analyze

Solution:

$P(X \cap D) = 0.70 \times 0.01 = 0.007$; $P(Y \cap D) = 0.30 \times 0.04 = 0.012$.

Total $P(D) = 0.007 + 0.012 = 0.019$.

$P(X|D) = 0.007 / 0.019 =$ **0.3684 (36.84%)**.

Q30. Apply Bayesian updating: Prior P(Fair coin) = 0.6, P(Biased coin) = 0.4 (where P(H|Biased)=0.75). Two consecutive heads are observed. Find posterior probability of biased coin.

Syllabus: Bayesian Inference | CO3 | Bloom L3 | Difficulty: Medium | Start word: Apply

Solution:

Likelihoods: $P(HH \mid Fair) = (0.5)^2 = 0.25$; $P(HH \mid Biased) = (0.75)^2 = 0.5625$.

Unnormalized posteriors: Fair = 0.6 × 0.25 = 0.15; Biased = 0.4 × 0.5625 = 0.225. Total = 0.375.

$P(Biased \mid HH) = 0.225 / 0.375 =$ **0.60 (60%)**.

Q31. Evaluate a hypothesis test for sample mean $\bar{x} = 104$, $\mu_0 = 100$, $s = 12$, $n = 25$ at 5% level (critical $t_{0.025,24} = 2.064$).

Syllabus: Frequentist Inference | CO3 | Bloom L5 | Difficulty: Difficult | Start word: Evaluate

Solution:

SE = 12 / √25 = 2.4. Test statistic $t = (104 - 100) / 2.4 = 1.667$.

Since $|t| = 1.667 < 2.064$, **Fail to reject H_0**.

95% Confidence Interval = 104 ± 2.064(2.4) = 104 ± 4.95 = **(99.05, 108.95)**.

Q32. Construct the least-squares regression line for data points: (1, 3), (2, 5), (3, 6), (4, 8), (5, 8). Predict y at x = 6.

Syllabus: Regression Analysis | CO3 | Bloom L6 | Difficulty: Difficult | Start word: Construct

Solution:

$\bar{x} = 3$, $\bar{y} = 6$. $S_{xx} = \sum(x - \bar{x})^2 = 4+1+0+1+4 = 10$.

$S_{xy} = \sum(x - \bar{x})(y - \bar{y}) = (-2)(-3) + (-1)(-1) + 0 + (1)(2) + (2)(2) = 6 + 1 + 0 + 2 + 4 = 13$.

Slope $b1 = 13 / 10 = 1.3$; Intercept $b0 = 6 - 1.3(3) = 2.1 \Rightarrow$ **$\hat{y} = 2.1 + 1.3x$**.

Prediction at x = 6: $\hat{y}(6) = 2.1 + 1.3(6) =$ **9.9**.

Unit IV: Graph Theory

Maps to CO4: Analyze graph theory concepts to solve practical problems.

Q33. State the total number of edges in a graph with degree sequence (4, 4, 3, 3, 2) using the Handshaking Lemma.

Syllabus: Graph Basics | CO4 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

Sum of degrees = 4 + 4 + 3 + 3 + 2 = 16.

By Handshaking Lemma: $2|E| = 16 \Rightarrow |E| =$ **8 edges**.

Q34. State the number of edges in K_7 and complete bipartite graph $K_{3,5}$.

Syllabus: Graph Types | CO4 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

K_7 edges = $n(n-1)/2 = (7 \times 6)/2 =$ **21**.

$K_{3,5}$ edges = $m \times n = 3 \times 5 =$ **15**.

Q35. Illustrate the vertex degrees and check Eulerian condition for a graph on {A, B, C, D} with edges AB, BC, CD, DA, AC.

Q37. Calculate the chromatic number for cycle graphs C_6, C_7, and complete graph K_5.

Syllabus: Graph Colouring Problem | CO4 | Bloom L3 | Difficulty: Easy | Start word: Calculate

> **Solution:**
> $\chi(C_6)$ = **2** (even cycle), $\chi(C_7)$ = **3** (odd cycle), $\chi(K_5)$ = **5** (complete graph).

Q38. Compute shortest path from S to all nodes in weighted graph: S→A=3, S→B=5, A→B=1, A→C=6, B→C=2.

Syllabus: Graph Searching Algorithms | CO4 | Bloom L3 | Difficulty: Medium | Start word: Compute

> **Solution:**
> d(S)=0; d(A)=3; d(B) = min(5, 3+1) = 4; d(C) = min(3+6, 4+2) = 6.
> Shortest distances: **S:0, A:3, B:4, C:6**. Path to C: S → A ⇒ B ⇒ C.

Q39. Analyze cut capacity separating {s} from rest with outgoing edges s→u (cap 10) and s→v (cap 7).

Syllabus: Max-Flow Min-Cut Algorithm | CO4 | Bloom L4 | Difficulty: Medium | Start word: Analyze

> **Solution:**
> Cut capacity = 10 + 7 = **17**. By Max-Flow Min-Cut Theorem, maximum flow cannot exceed 17.

Q40. Apply Dijkstra's Algorithm on graph from A: A→B=2, A→C=5, B→D=4, C→D=1, D→E=3.

Syllabus: Graph Searching Algorithms | CO4 | Bloom L3 | Difficulty: Medium | Start word: Apply

> **Solution:**
> Distances: d(A)=0, d(B)=2, d(C)=5. Reach D via B (2+4=6) or C (5+1=6) ⇒ d(D)=6. Reach E via D (6+3=9) ⇒ d(E)=9.
> Final Distances: **A:0, B:2, C:5, D:6, E:9**. Finalization order: A, B, C, D, E.

Q41. Examine DFS discovery and finish timestamps for DAG: A→B, A→C, B→D, C→D (Alphabetical order).

Syllabus: Graph Traversing Algorithms | CO4 | Bloom L4 | Difficulty: Difficult | Start word: Examine

> **Solution:**
> DFS Traversal: A(d:1) → B(d:2) → D(d:3, f:4) ⇒ B(f:5) ⇒ C(d:6, f:7) ⇒ A(f:8).
> Timestamps [d/f]: **A[1/8], B[2/5], C[6/7], D[3/4]**.
> Topological Order (decreasing finish time): **A, C, B, D**.

Q42. Evaluate Max-Flow in network: s→a=7, s→b=6, a→t=5, b→t=5, a→b=2.

Syllabus: Max-Flow Min-Cut Algorithm | CO4 | Bloom L5 | Difficulty: Difficult | Start word: Evaluate

> **Solution:**
> Path 1: s → a → t (flow 5, bottle neck 5). Path 2: s → b → t (flow 5, bottle neck 5).
> No more augmenting paths exist. **Max Flow = 10**. Min-Cut separates {s, a, b} and {t} with capacity 5 + 5 = 10.

Q43. Design Welsh-Powell graph coloring schedule for exams {A, B, C, D} with conflicting pairs A-B, A-C, B-C, C-D.

Syllabus: Graph Colouring Problem | CO4 | Bloom L6 | Difficulty: Difficult | Start word: Design

> **Solution:**
> Degrees: C:3, A:2, B:2, D:1. Order: C, A, B, D.
> Assign Color 1: C. Assign Color 2: A, D (since non-adjacent). Assign Color 3: B.
> **3 Time Slots needed**: Slot 1: {C}, Slot 2: {A, D}, Slot 3: {B}.

Unit V: Algorithm Complexity Analysis

Maps to CO5: Determine algorithmic complexity of mathematical problems.

Q44. State maximum comparisons binary search makes on a sorted array of 512 elements.

Syllabus: Searching and Sorting Problem | CO5 | Bloom L1 | Difficulty: Easy | Start word: State

> **Solution:**
> Worst case = $\lfloor \log_2 n \rfloor + 1 = \lfloor \log_2 512 \rfloor + 1 = 9 + 1 =$ **10 comparisons**.

Q47. Illustrate growth rates of n^2, 2^n, $n \log_2 n$ evaluated at n = 16.

Syllabus: Asymptotic Notations | CO5 | Bloom L2 | Difficulty: Easy | Start word: Illustrate

Solution:

$n \log_2 n = 16(4) = 64$ • $n^2 = 256$ • $2^n = 65{,}536$.

Increasing order: **$n \log_2 n < n^2 < 2^n$**.

Q48. Solve T(n) = T(n−1) + 3 with T(1) = 2 by iteration, and find T(40).

Syllabus: Recurrence Relations | CO5 | Bloom L3 | Difficulty: Medium | Start word: Solve

Solution:

Iteration: $T(n) = T(1) + 3(n-1) = 2 + 3n - 3 = 3n - 1$.

For n = 40: $T(40) = 3(40) - 1 =$ **119**.

Q49. Determine complexity of $T(n) = 8T(n/2) + n^2$ using Master Theorem.

Syllabus: Master Theorem | CO5 | Bloom L3 | Difficulty: Medium | Start word: Determine

Solution:

$a = 8, b = 2 \Rightarrow n^{\log_2 8} = n^3$.

$f(n) = n^2 = O(n^{3-\varepsilon})$ for $\varepsilon=1 \Rightarrow$ Case 1 applies $\Rightarrow$ **$T(n) = \Theta(n^3)$**.

Q50. Calculate recursion depth and worst-case comparisons for Merge Sort on 32 elements.

Syllabus: Searching and Sorting Problem | CO5 | Bloom L3 | Difficulty: Medium | Start word: Calculate

Solution:

Recursion tree levels = $\log_2(32) + 1 =$ **6 levels**.

Max comparisons ($n \log_2 n - n + 1$) = $32(5) - 32 + 1 = 160 - 31 =$ **129 comparisons**.

Q51. Analyze recurrence $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 1$, $a_1 = 4$. Find closed form.

Syllabus: Recurrence Relations | CO5 | Bloom L4 | Difficulty: Difficult | Start word: Analyze

Solution:

Characteristic equation: $r^2 - 5r + 6 = 0 \Rightarrow (r-2)(r-3) = 0 \Rightarrow r = 2, 3$.

General form: $a_n = A(2^n) + B(3^n)$.

$a_0 = A + B = 1$, $a_1 = 2A + 3B = 4 \Rightarrow B = 2, A = -1$.

Closed form: **$a_n = 2(3^n) - 2^n$**.

Q52. Construct truth table for 3-SAT CNF formula (x1 ∨ x2) ∧ (¬x1 ∨ ¬x2). Determine number of satisfying assignments.

Syllabus: Satisfiability Problem | CO5 | Bloom L6 | Difficulty: Difficult | Start word: Construct

Solution:

4 rows for (x1, x2): (F, F)→Clause 1 False; (T, T)→Clause 2 False.

Satisfying assignments: **(F, T) and (T, F) ⇒ 2 satisfying assignments out of 4**.

Unit VI: Algorithmic and Optimization Approach

Maps to CO6: Discuss different optimization approaches to solve mathematical problems.

Q53. State value-to-weight ratios for items A (w:4, v:24), B (w:8, v:32), C (w:10, v:50) and greedy sorting order.

Syllabus: Greedy Approach | CO6 | Bloom L1 | Difficulty: Easy | Start word: State

Solution:

Ratios: A = 24/4 = 6.0, B = 32/8 = 4.0, C = 50/10 = 5.0.

Greedy Selection Order: **A, C, B**.

Q56. Compute maximum value for Fractional Knapsack capacity W = 15, items A(w:5, v:30), B(w:10, v:50), C(w:3, v:12).

Syllabus: Greedy Approach | CO6 | Bloom L3 | Difficulty: Medium | Start word: Compute

> **Solution:**
> Ratios: A = 6.0, B = 5.0, C = 4.0.
> Take all A (w:5, v:30), take 10 units of B (w:10, v:50) ⇒ Total Weight = 15.
> Optimal Total Value = 30 + 50 = **80**.

Q57. Calculate Longest Common Subsequence (LCS) length for strings X = "ABACD" and Y = "BCDA".

Syllabus: Dynamic Programming | CO6 | Bloom L3 | Difficulty: Medium | Start word: Calculate

> **Solution:**
> Matching sub-sequences include "BCD", "ACD".
> Maximum LCS length = **3** (e.g., "BCD").

Q58. Determine maximum subarray sum for array [-3, 4, -1, 2, 1, -5, 4] using Kadane's algorithm.

Syllabus: Divide and Conquer | CO6 | Bloom L3 | Difficulty: Medium | Start word: Determine

> **Solution:**
> Running sum tracking: 4, 3, 5, 6, 1, 5. Maximum sum encountered = **6** (achieved by subarray [4, −1, 2, 1]).

Q59. Apply 0/1 Knapsack dynamic programming for capacity W = 6, items w = {2, 3, 4}, v = {12, 20, 30}. Find optimal value.

Syllabus: Dynamic Programming | CO6 | Bloom L3 | Difficulty: Medium | Start word: Apply

> **Solution:**
> Combinations within weight 6: {2,4}→w:6, v:42; {3,4}→w:7(exceeds); {2,3}→w:5, v:32.
> Optimal selection: Items 1 and 3 (weights 2 + 4 = 6) ⇒ Max Value = **42**.

Q60. Examine backtrack decision tree leaves for Subset-Sum set S = {3, 5, 7, 10} with target T = 12.

Syllabus: Backtracking | CO6 | Bloom L4 | Difficulty: Medium | Start word: Examine

> **Solution:**
> Total subset tree leaves = 2^4 = **16 leaves**.
> Valid subsets summing to 12: **{5, 7}** (1 solution).

Q61. Evaluate Huffman Coding for frequencies A:8, B:12, C:15, D:20, E:35. Compute total encoded bits.

Syllabus: Greedy Approach | CO6 | Bloom L5 | Difficulty: Difficult | Start word: Evaluate

> **Solution:**
> Tree Merges: (A+B)=20 ⇒ (C+20)=35 ⇒ (D+35)=55 ⇒ (E+55)=90.
> Code Lengths: E:2, D:2, C:3, A:4, B:4.
> Total bits = 35(2) + 20(2) + 15(3) + 8(4) + 12(4) = 70 + 40 + 45 + 32 + 48 = **235 bits**.

Q62. Design 1 generation of Genetic Algorithm maximizing f(x) = x^2 on 4-bit chromosomes: 0110(6), 1001(9), 1100(12), 0011(3).

Syllabus: Genetic Algorithm | CO6 | Bloom L6 | Difficulty: Difficult | Start word: Design

> **Solution:**
> Fitness values: f(6)=36, f(9)=81, f(12)=144, f(3)=9. Total Fitness = 270. Average = 67.5.
> Selection Probabilities: P(6)=0.133, P(9)=0.300, P(12)=0.533, P(3)=0.033.
> Crossover 1001 and 1100 after bit 2 ⇒ Offspring: **1000 (8 → f=64)** and **1101 (13 → f=169)**.
> Max fitness increased from 144 to **169**.